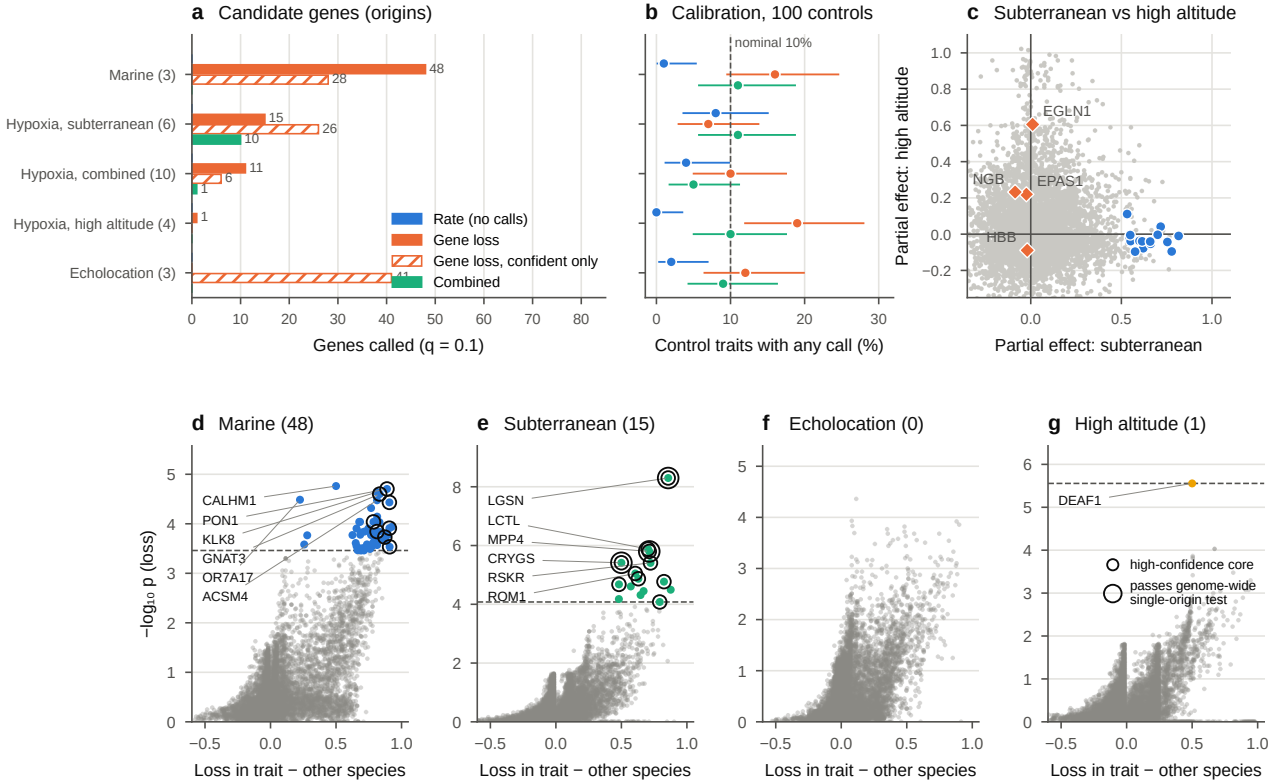

a Candidate genes (origins)
Marine (3)
Hypoxia, subterranean (6)
Hypoxia, combined (10)
Hypoxia, high altitude (4)
Echolocation (3)
48
28
15
26
10
11
6
1
1
41
Rate (no calls)
Gene loss
Gene loss, confident only
Combined
0 10 20 30 40 50 60 70 80
Genes called (q = 0.1)
b Calibration, 100 controls
nominal 10%
0 10 20 30
Control traits with any call (%)
c Subterranean vs high altitude
Partial effect: high altitude
1.0 0.8 0.6 0.4 0.2 0.0 −0.2
EGLN1
NGB
EPAS1
HBB
0.0 0.5 1.0
Partial effect: subterranean
d Marine (48)
−log10 p (loss)
CALHM1
PON1
KLK8
GNAT3
OR7A17
ACSM4
−0.5 0.0 0.5 1.0
Loss in trait − other species
e Subterranean (15)
LGSN
LCTL
MPP4
CRYGS
RSKR
RQM1
Loss in trait − other species
f Echolocation (0)
Loss in trait − other species
g High altitude (1)
DEAF1
high-confidence core
passes genome-wide single-origin test
Loss in trait − other species